

INSPECTION CERTIFICATE - REVISION 0.0

This document complies to BS EN 10204 3.1

CAST NO : 9316704

ORDER NO / CERT NO: 18NL 1078622



NAME AND ADDRESS OF SENDER

ArcelorMittal South Africa Limited 1989/002164/06
Long Steel Products
Leeuwkuil Mill Mario Milani Drive, Vereeniging Works

CUSTOMER ORD NO : P00102

CUSTOMER NO : 13622

SHIP :

SAP ORD NO : 1500413724

CUSTOMER

NAME AND ADDRESS OF CONSIGNEE

DESCRIPTION OF GOODS

SPRINGMASTER (PTY) LTD
P.O.Box 435
MOUNT EDGECOMBE
4300
South Africa

SPRINGMASTER DBN
136 OLD NORTH COAST ROAD
GLEN ANIL. DURBAN

Profile: Round Bar Lengths : 24.5mm

Specification: 51CrV4 (EN10089)

LENGTH : 6105.0 mm CONDITION: AS ROLLED

TOLERANCE : SS10/301

NO OF TESTS: 1

CHEMICAL ANALYSIS

	C %	Si %	Mn %	S %	P %	Cr %	Ni %	Mo %	Cu %	V %	Al %	Ti %	Sn %	Nb %	N %	B %	Ca %	As %	Sb %	Pb %	H ppm	CEV	Theoretical Mass (kg)	Actual Mass (kg)
Actual	0.54	0.31	0.95	0.004	0.011	1.110	0.110	0.035	0.140	0.127	0.006				0.005	0.0003					2.2		19543	19518
Minimum	0.47		0.70			0.900				0.100														
Maximum	0.55	0.40	1.10	0.025	0.025	1.200	0.300	0.150		0.250	0.050					0.0007								

JOMINY TEST (HRC)

	SCALE	1.5	3	5	7	8	9	10	11	12	13	15	17	19	20	25	30	35	40	45	50	
Actual (Ave)	mm	63	63	63	63		63		63		62	60			59	58	57	56	53	51	50	
Minimum																						
Maximum																						

MECHANICAL PROPERTIES

	Test Piece Direction	Test Temp °C	So mm ²	ReH MPa	Rm MPa	Std Dev Rm MPa	A %	Z %	Bend	Mass/ Meter Kg/m	Hardness HBW		R11	
											Low	High		
Actual (Ave)	L	25.00		864.0	1097.0		18.7	56.0			302.0	302.0	0.220	
Minimum														
Maximum														

METALLURGICAL RESULTS

	Grain Size ASTM E112	Decarburization				Inclusion ASTM type A										
		Partial		Complete		Sulphide (A)		Alumina (B)		Silicate (C)		Globular oxide (D)				
		Ave (mm)	Ave (%)	Max (mm)	Max (%)	Max (mm)	Max (%)	Thin	Thick	Thin	Thick	Thin	Thick		Thin	Thick
Actual	6.0	0.15	0.61	0.18	0.73			0.5	0.0	0.0	0.0	0.0	0.0	1.0	0.0	
Minimum	5.0															
Maximum	8.0		1.50		1.50			2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0	

ReH: Yield Stress Rm: Tensile Strength A: Elongation = 5.65*sqrt(So) Z: Reduction of Area So: Cross Sectional Area HRB: Rockwell hardness B scale HRC: Rockwell hardness C scale HBW: Brinell Hardness 10/3000 scale CEV: Carbon Equivalent Value R11 = Cu + 10Sn

STANDARD REMARKS :

Material 100% surface inspected to specification. 100% Spectrographically tested
Material Ultrasonically Tested and Approved - Pulse Echo A Scan Metho

S. WICKS
QUALITY MANAGEMENT

Issue Date: 2021-08-12

PRODUCT MANUFACTURED IN ACCORDANCE TO SPECIFICATION AND MET THE REQUIREMENTS

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End of Report